Build Quantum Mechanics from Scratch

1 Introduction

We derive the Schrödinger's equation of quantum mechanics from several basic experimental facts. These facts are treated as axioms. In fact, what we will get is a generalized Schrödinger equation, which is the most generic equation that these axioms can imply.

1.1 Conventions

- **Definitions** are in bold font.
- *Important* statements are in italic font.
- Only important equations are numbered.
- Questions are in red color.
- Conclusions are in green color.

1.2 Abbreviations

Because we are doing high-dimensional calculus, abbreviations are essential for not falling in the "debauchery of formulae".

When $x, y \in \mathbb{R}^d$, denote $xy = x \cdot y = \sum_\alpha x^\alpha y^\alpha$. Thus, $x^2 = x \cdot x = \sum_\alpha (x^\alpha)^2$. Furthermore, we use multi-dimensional index $\alpha = (\alpha_1, \ldots, \alpha_n)$ for abbreviating partial derivative $(\partial_{\alpha_1} \cdots \partial_{\alpha_n} f)(x)$ to $\partial_\alpha^n f(x)$, and abbreviating tensor (generalization of vector to multiple indices) $A^{\alpha_1 \cdots \alpha_n}$ to A^α, and $B_{\beta_1 \cdots \beta_n}$ to B_β. For example, the product $v^{\alpha_1} \times \cdots \times v^{\alpha_n}$ can be abbreviated as v^α.

Einstein's convention of summation is an elegant abbreviation for high-dimensional calculus and tensor analysis. It neglects the Σ_α in $\sum_\alpha v_\alpha w^\alpha$, abbreviated to be $v_\alpha w^\alpha$. The indices are "balanced", one superscript, and the other subscript. The convention does not apply to the "unbalanced" summation such as $\sum_\alpha v^\alpha w^\alpha$, because both indices are superscripts.

2 Probability Interpretation and Wavefunction

In classical Newtonian world, the states of a physical system with n particles are represented by elements in $\mathbb{R}^{6n}$, or **phases**, $3n$ dimensions for positions and $3n$ for velocities (or momenta). The time evolution is thus a trajectory in the phase space, namely a map $\mathbb{R} \to \mathbb{R}^{6n}$ (with time as the domain).

But for a quantum system, the states are represented by wavefunctions. A **wavefunction** is a map from the positions of particles, together with time, to complex plane. Thus, a wavefunction maps $\mathbb{R}^d \times \mathbb{R} \to \mathbb{C}$ where $d = 3n$. The first axiom claims how wavefunctions characterize a quantum system.

Axiom 1. *[Probability Interpretation] The states of a quantum system are represented by wavefunctions. And given a wavefunction φ, the probabilistic density that the particles are found in positions x at time t is given by $|\varphi(x,t)|^2 = \varphi^*(x,t)\varphi(x,t)$.*

3 Superposition Principle and Time Evolution

The second axiom of quantum mechanics, superposition principle, claims that operations on wavefunctions shall be linear (so that wavefunctions can be superpositioned).

Axiom 2. *[Superposition Principle] Physical laws that operate on quantum states shall be linear.*

An implication of superposition principle is how quantum states (precisely, their wavefunctions) evolve with time. Axiom 2 implies that the equation of time evolution (as a physical law that operates on a quantum state) shall be linear: $\partial\varphi/\partial t = L(\varphi)$ where the operation L is linear. Mathematically, linearity imports a kernel $r: \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{C}$ such that

$$\mathrm{i}\frac{\partial\varphi}{\partial t}(x,t) = \int_{\mathbb{R}^d} \mathrm{d}y\, r(x,y)\varphi(y,t). \tag{1}$$

The right hand side can be seen as a generalization of linear transformation in $\mathbb{R}^n$ like vector-matrix product $\sum_j r_{ij}\varphi_j$. The imaginary i is employed for convenience.

A direct result of axiom 1 is that probabilistic density shall be normalized. Namely, for any wave-function φ and any $t \in \mathbb{R}$, we shall have

$$\int_{\mathbb{R}^d} \mathrm{d}x\, \varphi^*(x,t)\varphi(x,t) = 1. \tag{2}$$

Taking derivative on t gives

$$\int_{\mathbb{R}^d} \mathrm{d}x\, \frac{\partial\varphi^*}{\partial t}(x,t)\varphi(x,t) + \int_{\mathbb{R}^d} \mathrm{d}x\, \varphi^*(x,t)\frac{\partial\varphi}{\partial t}(x,t) = 0.$$

Replacing $(\partial\varphi/\partial t)$ by equation 1 (and its complex conjugation),

$$\int_{\mathbb{R}^d} \mathrm{d}x \int_{\mathbb{R}^d} \mathrm{d}y \varphi(x,t)[r^*(x,y) - r(y,x)]\varphi^*(y,t) = 0.$$

Since φ is arbitrary, we obtain

$$r^*(x,y) = r(y,x). \tag{3}$$

That is, complex conjugating r is simply swapping its arguments. We call such function **Hermitian**. The two arguments of r are not independent.

Probability interpretation (axiom 1), together with superposition principle (axiom 2), is the direct result of the double-slit experiment of electron. Details can be found in Feynman's Lectures on Physics, Vol 3, chapter 1.

4 Wavefunction Is Rapidly Decreasing and Entire

Before calculations, we first address where wavefunctions live. Probability interpretation (axiom 1) demands that wavefunctions are square-integrable. Namely, wavefunction is in the square-integrable space $L^2(\mathbb{R}^d)$. But this is far from sufficient. Many mathematical tools are essential for evaluating quantum mechanics. Two of the many are Fourier transform and Taylor series.

It is well known, however, that the Fourier transform of a square-integrable function may not be square-integrable again, so that its inverse Fourier transform may not exist. Since Fourier transform is basic in quantum mechanics, we shall seek for a smaller space in which wavefunctions live. An ideal substitution is the (complex) rapidly decreasing functions. **Rapidly decreasing function** is a smooth function $f: \mathbb{R}^d \to \mathbb{C}$ that decays "exponentially fast" at infinity. Precisely, for any m-order polynomial P_m and any m-order partial derivative ∂^n,[1] with integers $m, n \geqslant 0$, we have

$$\lim_{\|x\| \to \infty} |P_m(x)\partial^n f(x)| = 0.$$

1. For example, $\partial_\alpha \partial_\beta^2$ is 3-order partial derivative, and $\partial_\alpha \partial_\beta^2 \partial_\gamma^9$ is 12-order.

Fourier transform on a rapidly decreasing function results in a rapidly decreasing function.[2] Further, a function that is *everywhere* consistent with its Taylor series is **entire** (shall not confuse with the entire function in complex analysis, where the domain is complex space). It indicates that we shall further restrict wavefunctions to be entire. Denote $\mathcal{S}_E(\mathbb{R}^d)$ as the collection of all the rapidly decreasing entire functions.

For example, in traditional textures, solving the stationary Schrödinger's equation of one-dimensional harmonic oscillator needs the expansion of wavefunction

$$\varphi(x) = \exp(-x^2)\left[\sum_{n=0}^{\infty} a_n x^n\right],$$

where the coefficients $(a_0, a_1, \ldots)$ are to be determined. The factor $\exp(-x^2)$ is employed for an exponentially fast decay as $|x| \to \infty$. And the factor $[\cdots]$ is a Taylor series. Hence, f is both rapidly decreasing and entire. The energy quantization emerges for ensuring the convergence of the series $\sum_n a_n x^n$.

Rapidly decreasing entire functions are dense in square-integrable space, meaning that for any $f \in L^2(\mathbb{R}^d)$ and any $\varepsilon > 0$, there is a $g \in \mathcal{S}_E(\mathbb{R}^d)$ such that

$$\int_{\mathbb{R}^d} \mathrm{d}x\, |f(x) - g(x)|^2 < \varepsilon.$$

For example, for any wavefunction $f \in L^2(\mathbb{R}^d)$, when we measure the probability on any area of positions $U \subset \mathbb{R}^d$, we can use its approximation $g \in \mathcal{S}_E(\mathbb{R}^d)$ instead, because the difference is bounded by

$$\left|\int_U \mathrm{d}x |f(x)|^2 - \int_U \mathrm{d}x |g(x)|^2\right| \leqslant \int_U \mathrm{d}x\, |f(x) - g(x)|^2 \leqslant \int_{\mathbb{R}^d} \mathrm{d}x\, |f(x) - g(x)|^2 < \varepsilon.$$

The first inequality is equivalent to $|\|f\| - \|g\|| \leqslant \|f - g\|$, where the norm is defined as $\|f\| := \int_U \mathrm{d}x |f(x)|^2$, recognized as the L^2-norm on U. It states that the difference between the two sides of a triangle is less than that of the third side. It is in this sense that the substitution $L^2(\mathbb{R}^d) \to \mathcal{S}_E(\mathbb{R}^d)$ is plausible. Proof of the statement that rapidly decreasing entire functions are dense in square-integrable space is given in appendices B.

5 Path Integral Formalism

We are trying to derive a generic path integral formalism. Given a small $\Delta t > 0$, equation 1 gives

$$\varphi(x, t + \Delta t) = \int_{\mathbb{R}^d} \mathrm{d}y [\delta(x - y) - \mathrm{i} r(x, y) \Delta t]\, \varphi(y, t) + o(\Delta t).$$

We are to convert the $[\cdots]$ part into exponential. To do so, we take the inverse Fourier transform

$$\delta(x - y) = \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d} \exp(\mathrm{i}k(x - y)),$$

and

$$r(x, y) = \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d} \exp(\mathrm{i}k(x - y)) \hat{r}(k, y), \tag{4}$$

2. This was first found by Laurent Schwartz in 1949, in his paper "Théorie des distributions et transformation de Fourier". The space of rapidly decreasing functions is now called **Schwartz space**, denoted by $\mathcal{S}(\mathbb{R}^d)$. Thus, Fourier transform is a linear automorphism in Schwartz space.

in which[3] [4]

$$\hat{r}(k,y) := \int_{\mathbb{R}^d} \mathrm{d}x \exp(-\mathrm{i}k(x-y))\, r(x,y). \tag{5}$$

Then, the $[\cdots]$ part is converted into exponential by

$$\begin{aligned}[\cdots] &= \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d} \exp(\mathrm{i}k(x-y))[1 - \mathrm{i}\hat{r}(k,y)\Delta t] \\ &= \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d} \exp\{\mathrm{i}k(x-y) - \mathrm{i}\hat{r}(k,y)\Delta t\} + o(\Delta t)\end{aligned}$$

Plugging back to $\varphi(x, t+\Delta t)$ results in

$$\varphi(x, t+\Delta t) = \int_{\mathbb{R}^d} \mathrm{d}y \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d} \exp\{\mathrm{i}k(x-y) - \mathrm{i}\hat{r}(k,y)\Delta t\} \varphi(y,t) + o(\Delta t).$$

Now, we have converted the $[\cdots]$ part into exponential, as a starting point of constructing path integral.

After re-denoting $x_1 := x$, $x_0 := y$, $k_0 := k$, it becomes

$$\varphi(x_1, t+\Delta t) = \int_{\mathbb{R}^d} \mathrm{d}x_0 \int_{\mathbb{R}^d} \frac{\mathrm{d}k_0}{(2\pi)^d} \exp\{\mathrm{i}k_0(x_1 - x_0) - \mathrm{i}\hat{r}(k_0, x_0)\Delta t\}\, \varphi(x_0, t) + o(\Delta t).$$

The same (replacing x_1 by x_2, and x_0 by x_1),

$$\varphi(x_2, t+2\Delta t) = \int_{\mathbb{R}^d} \mathrm{d}x_1 \int_{\mathbb{R}^d} \frac{\mathrm{d}k_1}{(2\pi)^d} \exp\{\mathrm{i}k_1(x_2 - x_1) - \mathrm{i}\hat{r}(k_1, x_1)\Delta t\}\, \varphi(x_1, t+\Delta t) + o(\Delta t).$$

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3. Indeed, by inserting equation 5 into equation 4, we get

$$r(x,y) = \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d} \exp(\mathrm{i}k(x-y)) \times \left[\int_{\mathbb{R}^d} \mathrm{d}x' \exp(-\mathrm{i}k(x'-y))\, r(x',y)\right].$$

Re-arrange the right hand side as

$$\int_{\mathbb{R}^d} \mathrm{d}x'\, r(x',y) \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d} \exp(\mathrm{i}k(x-x')) = \int_{\mathbb{R}^d} \mathrm{d}x'\, r(x',y) \delta(x-x'),$$

which goes back to $r(x,y)$, indicating that equations 4 and 5 are consistent.

4. Alternatively, we can define

$$\tilde{r}(x,k) := \int_{\mathbb{R}^d} \mathrm{d}y \exp(-\mathrm{i}k(x-y))\, r(x,y).$$

Thus,

$$r(x,y) = \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d} \exp(\mathrm{i}k(x-y)) \tilde{r}(x,k).$$

Indeed, by plugging $\tilde{r}(x,k)$ into the right hand side of $r(x,y)$,

$$r(x,y) = \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d} \exp(\mathrm{i}k(x-y)) \times \left[\int_{\mathbb{R}^d} \mathrm{d}y' \exp(-\mathrm{i}k(x-y'))\, r(x,y')\right].$$

Re-arrange the right hand side as

$$\int_{\mathbb{R}^d} \mathrm{d}y'\, r(x,y') \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d} \exp(\mathrm{i}k(y'-y)) = \int_{\mathbb{R}^d} \mathrm{d}y'\, r(x,y') \delta(y'-y),$$

which goes back to $r(x,y)$ again. The $\hat{r}$ and $\tilde{r}$ are the Fourier transform of r performed on each of its arguments respectively. In fact, $\hat{r}$ and $\tilde{r}$ are the same. Indeed, recalling the Hermitianity of r, we have

$$\hat{r}^*(k,y) = \int_{\mathbb{R}^d} \mathrm{d}x \exp(\mathrm{i}k(x-y))\, r^*(x,y) = \int_{\mathbb{R}^d} \mathrm{d}x \exp(\mathrm{i}k(x-y)) r(y,x).$$

Exchanging x and y makes

$$\hat{r}^*(k,x) = \int_{\mathbb{R}^d} \mathrm{d}y \exp(-\mathrm{i}k(x-y)) r(x,y),$$

which is just the $\tilde{r}(x,k)$. So, we have

$$\hat{r}^*(k,x) = \tilde{r}(x,k).$$

Once again, we find that the two arguments of $r(x,y)$ are not independent.

By inserting $\varphi(x_1, t + \Delta t)$, we find

$$\begin{aligned}\varphi(x_2, t + 2\Delta t) = &\int_{\mathbb{R}^d} \mathrm{d}x_1 \int_{\mathbb{R}^d} \frac{\mathrm{d}k_1}{(2\pi)^d} \exp\{\mathrm{i}k_1(x_2 - x_1) - \mathrm{i}\hat{r}(k_1, x_1)\Delta t\} \times \\ &\times \int_{\mathbb{R}^d} \mathrm{d}x_0 \int_{\mathbb{R}^d} \frac{\mathrm{d}k_0}{(2\pi)^d} \exp\{\mathrm{i}k_0(x_1 - x_0) - \mathrm{i}\hat{r}(k_0, x_0)\Delta t\}\, \varphi(x_0, t) \\ &+ o(\Delta t).\end{aligned}$$

After repeating this N times, we arrive at

$$\varphi(x_N, t + N\Delta t) = \int D(k, x) \exp(\mathrm{i}S(k, x)) \varphi(x_0, t) + o(\Delta t), \tag{6}$$

in which

$$\int D(k, x) := \prod_{i=0}^{N-1} \int_{\mathbb{R}^d} \frac{\mathrm{d}k_i}{(2\pi)^d} \int_{\mathbb{R}^d} \mathrm{d}x_i \tag{7}$$

and

$$S(k, x) := \sum_{i=0}^{N-1} \Delta t \Big[k_i \Big(\frac{x_{i+1} - x_i}{\Delta t}\Big) - \hat{r}(k_i, x_i)\Big]. \tag{8}$$

If we recognize $(x_{i+1} - x_i) / \Delta t$ as the velocity $\dot{x}(t_i)$, then $S(k, x)$ can be seen as the Legendre transform $\int [p(t)\dot{x}(t) - H(p(t), x(t))]\mathrm{d}t$, in which *$k$ is analogy to momentum p and $\hat{r}(k, x)$ plays the role of Hamiltonian $H(p, x)$*.

6 From Integral to Differential

Given a "test function" $\psi \in \mathcal{S}_E(\mathbb{R}^d)$, we can calculate its "inner product" with equation 1, as

$$\int_{\mathbb{R}^d} \mathrm{d}x \psi(x) \Big[\mathrm{i}\frac{\partial \varphi}{\partial t}(x, t)\Big] = \int_{\mathbb{R}^d} \mathrm{d}x \int_{\mathbb{R}^d} \mathrm{d}y \psi(x) r(x, y) \varphi(y, t).$$

In the right hand side, Taylor expanding ψ at y makes

$$\int_{\mathbb{R}^d} \mathrm{d}x \psi(x) \Big[\mathrm{i}\frac{\partial \varphi}{\partial t}(x, t)\Big] = \sum_{n=0}^{\infty} \frac{1}{n!} \int_{\mathbb{R}^d} \mathrm{d}x \int_{\mathbb{R}^d} \mathrm{d}y \partial_\alpha^n \psi(y) (x - y)^\alpha r(x, y) \varphi(y, t),$$

where $\alpha := (\alpha_1, \ldots, \alpha_n)$ and $(x - y)^\alpha := (x - y)^{\alpha_1} \times \cdots \times (x - y)^{\alpha_n}$ (recall the abbreviations). Define the **moment**

$$R_n^\alpha(y) := \int_{\mathbb{R}^d} \mathrm{d}x\, r(x, y) (x - y)^\alpha. \tag{9}$$

Then, integrating over x gives

$$\int_{\mathbb{R}^d} \mathrm{d}x \psi(x) \Big[\mathrm{i}\frac{\partial \varphi}{\partial t}(x, t)\Big] = \sum_{n=0}^{\infty} \frac{1}{n!} \int_{\mathbb{R}^d} \mathrm{d}y \partial_\alpha^n \psi(y) R_n^\alpha(y) \varphi(y, t).$$

After integration by parts and then omitting the boundary (since ψ is rapidly decreasing), we get

$$\int_{\mathbb{R}^d} \mathrm{d}y \psi(y) \Big[\mathrm{i}\frac{\partial \varphi}{\partial t}(y, t)\Big] = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_{\mathbb{R}^d} \mathrm{d}y \psi(y) \partial_\alpha^n [R_n^\alpha(y) \varphi(y, t)],$$

where we have replaced x by y in the left hand side for making it clear. Since ψ is an arbitrary function in $\mathcal{S}_E(\mathbb{R}^d)$, we have[5]

$$\mathrm{i}\frac{\partial \varphi}{\partial t}(x, t) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \partial_\alpha^n [R_n^\alpha(x) \varphi(x, t)]. \tag{10}$$

That is, we convert the integral equation 1 to a differential equation. In practice, differential equation is much more convenient than its integral correspondence for doing calculus.

5. This is a quantum analogy of the Kramers–Moyal expansion in stochastic process.

Interestingly, the Taylor expansion of the "Hamiltonian" $\hat{r}(k,y)$, defined by equation 5, also relates to the moments R_ns. Directly by equation 5, we have

$$\frac{\partial^n \hat{r}}{\partial k_\alpha}(0,y) = \lim_{k\to 0} \int_{\mathbb{R}^d} \mathrm{d}x \left[\frac{\partial^n}{\partial k_\alpha} \exp(-\mathrm{i}k(x-y))\right] r(x,y) = (-\mathrm{i})^n \int_{\mathbb{R}^d} \mathrm{d}x\, r(x,y)(x-y)^\alpha.$$

The integral is recognized as $R_n^\alpha(y)$. So, we find $(-\mathrm{i})^n R_n^\alpha(y)$s the Taylor coefficients of $\hat{r}(k,y)$ expanded by k at its origin. Namely,

$$\hat{r}(k,y) = \sum_{n=0}^{\infty} \frac{(-\mathrm{i})^n}{n!} R_n^\alpha(y) k_\alpha, \tag{11}$$

where $k_\alpha := (k_{\alpha_1} \times \cdots \times k_{\alpha_n})$ as usual. Again, the details of $S(k,x)$ (defined in equation 6) can be completely determined by the moments R_ns.

So, consider the traditional Hamiltonian $\hat{r}(x,p) = p^2/(2m) + V(x)$, all R_ns vanish except for $R_0(x) = V(x)$ and $R_2(x) = -1/m$. Equation 10 implies

$$\mathrm{i}\frac{\partial \varphi}{\partial t}(x,t) = -\frac{1}{2m}\nabla^2 \varphi(x,t) + V(x)\varphi(x,t), \tag{12}$$

which is exactly the Schrödinger equation (in the natural units where Planck's constant $\hbar = 1$).

7 Locality Truncates the Moments

We then introduce the third axiom about locality, and discuss what it will induce.

Axiom 3. *[Locality] Time evolution of quantum system is local.*

Shall not confuse time evolution with collapse, which is proven to be non-local. Axiom 3 claims that equation 1 is local. To make this clear, we consider an example, in which $R_n(x) = c^n$ for some constant c, and set the dimension $d = 1$ for simplicity. Then, time evolution (equation 10) at $x = 0$ is

$$\mathrm{i}\frac{\partial \varphi}{\partial t}(0,t) = \sum_{n=0}^{\infty} \frac{(-c)^n}{n!} \frac{\partial^n \varphi}{\partial x^n}(0,t).$$

The last expression happens to be the Taylor series of $\varphi(x,t)$ at $x = -c$, namely $\varphi(-c,t)$. So, we conclude that $R_n(x) = c^n$ for some constant c implies

$$\mathrm{i}\frac{\partial \varphi}{\partial t}(0,t) = \varphi(-c,t).$$

If we change the value of φ at $x = -c$, then the time evolution at $x = 0$ changes accordingly. It means non-locality.

In physics, a local equation generally refers to an operation on φ which contains φ itself and *finite* number of spatial derivatives of φ, such as

$$\mathrm{i}\frac{\partial \varphi}{\partial t}(x,t) = \mathcal{L}\left(\varphi(x,t), \frac{\partial \varphi}{\partial x}(x,t), \frac{\partial^2 \varphi}{\partial x^2}(x,t), \ldots, \frac{\partial^n \varphi}{\partial x^n}(x,t)\right),$$

where $\mathcal{L}$ is an entire function. This is easy to understand because to compute $(\partial^n \varphi/\partial x^n)(0,t)$ using numerical method with difference Δx, only $\varphi(x,t)$ with $x \in \{0, \Delta x, \ldots, n\Delta x\}$ are employed.[6] So, $(\partial \varphi/\partial t)(0,t)$ cannot "perceive" the $\varphi(x,t)$ outside the neighborhood $\{x: |x| \leqslant n\Delta x\}$. Since Δx can be arbitrarily small (but not vanishing), the equation is local.

We have shown that, with a cut-off N_{cut} on R_ns such that $R_n = 0$ for any $n > N_{\text{cut}}$, time evolution of wavefunction is local. And without such a cut-off, we can construct a sequence of R_ns such that time evolution is not local. Now, we are to prove that, generally, without a cut-off, any sequence of R_ns (such that for any $N > 0$, there are infinite many R_ns that are not vanishing), time evolution of wavefunction is non-local. This then imports a cut-off on moments.

6. Given a smooth function f, numerically evaluating $f^{(n)}(0)$ needs $f^{(n-1)}(0)$ and $f^{(n-1)}(\Delta x)$. Recursively, numerically evaluating $f^{(n-1)}(0)$ needs $f^{(n-2)}(0)$ and $f^{(n-2)}(\Delta x)$, and $f^{(n-1)}(\Delta x)$ needs $f^{(n-2)}(\Delta x)$ and $f^{(n-2)}(2\Delta x)$. Repeating this process, we find $\{f(0), \ldots, f(n\Delta x)\}$ are needed for evaluating $f^{(n)}(0)$.

Before proving, we first declare a property of entire function. Consider the Taylor expansion of $\varphi \in \mathcal{S}_E(\mathbb{R})$ at origin

$$\varphi(x) = a_0 + a_1 x + a_2 x^2 + \cdots.$$

The information of φ is completely encoded in the infinite sequence of a_ns. This is the result of a theorem which claims that two entire functions are equal if they agree in any interval (hence we can obtain the Taylor series of the function within the interval). What if we only know a portion of the infinite sequence of a_ns? For example, if we only know the a_0, then only the value of $\varphi(x)$ at $x = 0$ is determined. Further, if we also know the a_1, then we can give a good approximation to $\varphi(x)$ in a very tiny neighborhood at $x = 0$, since $\varphi(x) = a_0 + a_1 x + o(x^2)$. Then, if we additionally know the a_2, then the approximation becomes as good as before in a larger neighborhood at $x = 0$, since $\varphi(x) = a_0 + a_1 x + a_2 x^2 + o(x^3)$ and the size of neighborhood can increase a little without growing the scale of residue. This analysis indicates that the more a_ns we know, in a larger neighborhood at $x = 0$ can we properly approximate $\varphi(x)$. Our vision becomes border and border if we know more and more a_ns. Until knowing the whole sequence of a_ns, we realize the complete picture of $\varphi(x)$ (based on the previous theorem about entire function).

It also indicates that, for a sufficient large N, we can keep $\varphi(x)$ (approximately) invariant when we tune the a_ns with $n > N$ while keeping the other a_ns unchanged. On the other hand, based on the equation (plugging $\varphi(x) = \sum_n a_n x^n$ into equation 10, and collecting all a_m terms with $m < n$ into $[\cdots]$),

$$\mathrm{i}\frac{\partial \varphi}{\partial t}(0, t) = \sum_{n=0}^{\infty} \{[\cdots] + (-1)^n R_n(0) a_n\},$$

if there is not a cut-off to the infinite sequence of $R_n(0)$s, we can modify the $(\partial\varphi / \partial t)(0, t)$ by tuning an a_n where n can be arbitrarily large. This, however, will leave the $\varphi(x)$ around $x = 0$ invariant. It means that the value of φ with x far away from $x = 0$ can affect the time evolution of φ at $x = 0$. That is, time evolution of wavefunction is non-local. Hence, there must be a cut-off on the sequence of $R_n(0)$s. This discussion also holds for any value of x other than $x = 0$. We conclude that *time evolution is local if and only if there is a positive integer N_{cut} such that $R_n = 0$ for any $n > N_{\text{cut}}$*.

8 Hermitianity on the Moments of Transition Rate

The complex conjugation of moment is given by

$$(R_n^\alpha)^*(y) = \int_{\mathbb{R}^d} \mathrm{d}x\, r^*(x, y)(x - y)^\alpha = \int_{\mathbb{R}^d} \mathrm{d}x\, r(y, x)(x - y)^\alpha,$$

where we have inserted the Hermitian condition $r^*(x, y) = r(y, x)$. Exchanging x and y makes

$$(R_n^\alpha)^*(x) = \int_{\mathbb{R}^d} \mathrm{d}x\, r(x, y)(y - x)^\alpha = (-1)^n \int_{\mathbb{R}^d} \mathrm{d}y\, r(x, y)(x - y)^\alpha.$$

Remark that the integral is symmetric to $R_n^\alpha(y)$, which shares the same integrand (namely $r(x, y)(x - y)^\alpha$), but integrating over x instead. In equation 1, Taylor expanding $\varphi(y, t)$ by y at x gives

$$\mathrm{i}\frac{\partial \varphi}{\partial t}(x, t) = \int_{\mathbb{R}^d} \mathrm{d}y\, r(x, y) \varphi(y, t) = \sum_{n=0}^{\infty} \frac{1}{n!} \partial_\alpha^n \varphi(x, t) \int_{\mathbb{R}^d} \mathrm{d}y\, r(x, y)(y - x)^\alpha.$$

We recognize that the integral is nothing but the $(R_n^\alpha)^*(x)$. Hence,

$$\mathrm{i}\frac{\partial \varphi}{\partial t}(x, t) = \sum_{n=0}^{\infty} \frac{1}{n!} (R_n^\alpha)^*(x) \partial_\alpha^n \varphi(x, t). \tag{13}$$

Comparing with equation 10, this equation looks simpler.

Now we study the relation between R_ns and their complex conjugations. Direct calculation is found tedious. Instead, we start at evaluating $\hat{r}^*$. By conjugating equation 5 and applying the Hermitianity of r, we get

$$\hat{r}^*(k, y) = \int_{\mathbb{R}^d} \mathrm{d}x \exp(\mathrm{i}k(x - y))\, r(y, x).$$

Then, applying equation 4 to $r(y,x)$ gives

$$\hat{r}^*(k,y) = \int_{\mathbb{R}^d} \mathrm{d}k' \int_{\mathbb{R}^d} \frac{\mathrm{d}x}{(2\pi)^d} \exp\{\mathrm{i}(k-k')(x-y)\}\, \hat{r}(k',x).$$

Taylor expanding $\hat{r}(k',x)$ by x at y,

$$\hat{r}^*(k,y) = \sum_{n=0}^{\infty} \frac{1}{n!} \int_{\mathbb{R}^d} \mathrm{d}k' \frac{\partial^n \hat{r}}{\partial y^\alpha}(k',y) \times \int_{\mathbb{R}^d} \frac{\mathrm{d}x}{(2\pi)^d} \exp\{\mathrm{i}(k-k')(x-y)\}(x-y)^\alpha.$$

Notice the relation

$$\partial_\alpha^n \delta(\omega) = \frac{\partial^n}{\partial \omega^\alpha} \int_{\mathbb{R}^d} \frac{\mathrm{d}z}{(2\pi)^d} \exp(\mathrm{i}\omega z) = \mathrm{i}^n \int_{\mathbb{R}^d} \frac{\mathrm{d}z}{(2\pi)^d} \exp(\mathrm{i}\omega z) z^\alpha.$$

Replacing ω by $(k-k')$ and z by $(x-y)$, the last integral in $\hat{r}^*(k,y)$ becomes $(-\mathrm{i})^n \partial_\alpha^n \delta(k-k')$. Thus,

$$\hat{r}^*(k,y) = \sum_{n=0}^{\infty} \frac{(-\mathrm{i})^n}{n!} \int_{\mathbb{R}^d} \mathrm{d}k' \frac{\partial^n \hat{r}}{\partial y^\alpha}(k',y) \partial_\alpha^n \delta(k-k').$$

By integration by parts (recalling that $\partial^n \delta$ is odd when n is odd, otherwise even) and then integrating over k', we arrive at

$$\hat{r}^*(k,y) = \sum_{n=0}^{\infty} \frac{(-\mathrm{i})^n}{n!} \frac{\partial^{2n} \hat{r}}{\partial k_\alpha \partial y^\alpha}(k,y).$$

Since $(-\mathrm{i})^n R_n(y)$s are the coefficients of Taylor expansion of $\hat{r}(k,y)$ by k (equation 11), we find

$$(R_m^\alpha)^*(y) = (-\mathrm{i})^m \frac{\partial^m \hat{r}^*}{\partial k_\alpha}(0,y) = \sum_{n=0}^{\infty} \frac{(-\mathrm{i})^{m+n}}{n!} \frac{\partial^n}{\partial y^\beta} \frac{\partial^{m+n} \hat{r}}{\partial k_\alpha \partial k_\beta}(0,y).$$

Again, the $(m+n)$-th coefficients of the Taylor expansion is $(-\mathrm{i})^{m+n} R_{m+n}(y)$, so

$$(R_m^\alpha)^*(y) = \sum_{n=0}^{\infty} \frac{(-\mathrm{i})^{m+n}}{n!} \frac{\partial^n}{\partial x^\beta} [(-\mathrm{i})^{m+n} R_{m+n}^{\alpha\beta}(y)] = \sum_{n=0}^{\infty} \frac{(-1)^{m+n}}{n!} \partial_\beta^n R_{m+n}^{\alpha\beta}(x).$$

Recall that $R_n = 0$ for any $n > N_{\mathrm{cut}}$, we finally arrive at

$$(R_m^\alpha)^*(x) = \sum_{n=0}^{N_{\mathrm{cut}}-m} \frac{(-1)^{m+n}}{n!} \partial_\beta^n R_{m+n}^{\alpha\beta}(x). \tag{14}$$

It relates the moments R_ns to their complex conjugations.

For example, when $N_{\mathrm{cut}} = 2$, we have $R_0^*(x) = R_0(x) - \partial_\alpha R_1^\alpha(x) + (1/2)\partial_\alpha^2 R_2^\alpha(x)$, $(R_1^\alpha)^*(x) = -R_1^\alpha(x) + \partial_\beta R_2^{\alpha\beta}(x)$, and $(R_2^\alpha)^*(x) = R_2^\alpha(x)$, that is, R_2 is real. Specially, we have

$$R_{N_{\mathrm{cut}}}^*(x) = (-1)^{N_{\mathrm{cut}}} R_{N_{\mathrm{cut}}}(x).$$

That is, $R_{N_{\mathrm{cut}}}$ is real when N_{cut} is even, otherwise purely imaginary.

9 Uncertainty Principle Restricts the Highest Order Moment

An **observable** is a real function of the particle positions of a quantum system. A corollary of axiom 1 is that, given a wavefunction φ and an observable F, $|\varphi(x,t)|^2$ represents the probability density that the particles are found on positions x at time t, hence the observed value is the expectation

$$\mathbb{E}_\varphi[F](t) := \int_{\mathbb{R}^d} \mathrm{d}x\, |\varphi(x,t)|^2 F(x). \tag{15}$$

And the variance

$$\mathrm{Var}_\varphi[F](t) := \int_{\mathbb{R}^d} \mathrm{d}x\, |\varphi(x,t)|^2 [F(x) - \mathbb{E}_\varphi[F](t)]^2. \tag{16}$$

For a single particle system, we use X to denote the position of the particle, thus $X(x)=x$. Apparently, X is an observable. Its uncertainty is given by

$$\Delta X(t) := \sqrt{\mathrm{Var}_{\varphi}[X](t)}. \tag{17}$$

Having observed the particle position at time t and $t+\Delta t$ subsequently, we can obtain the velocity of the particle

$$V(t) := \lim_{\Delta t \to 0} \frac{\mathbb{E}_{\varphi}[X](t+\Delta t) - \mathbb{E}_{\varphi}[X](t)}{\Delta t}. \tag{18}$$

The uncertainty of velocity is hard to determine, because both ends (particle positions at time t and $t+\Delta t$) are uncertain. The key is fixing the position at time t by $\mathbb{E}_{\varphi}[X](t)$, and measuring the uncertainty caused by the position at time $t+\Delta t$. That is, *the uncertainty induced by time evolution*. Thus, we have

$$\Delta V(t) := \sqrt{\lim_{\Delta t \to 0} \mathrm{Var}_{\varphi}\left[\frac{X - \mathbb{E}_{\varphi}[X](t)}{\Delta t}\right](t+\Delta t)}. \tag{19}$$

Have clarified the uncertainties of particle position and velocity, we can claim uncertainty principle as follow.

Axiom 4. *[Uncertainty Principle of Single Particle] In a single particle quantum system, given a wavefunction φ, the uncertainties of position and of velocity have the relation*

$$\Delta X(t) \Delta V(t) \sim \hbar / (2m),$$

where m is the mass of the particle, and $\hbar$ is the reduced Planck's constant.

In the rest of this section, we investigate how uncertainty principle restricts further to the kernel. We first deal with the situation where dimension $d=1$. Since axiom 4 holds for all wavefunctions of a single particle, we choose φ to be Gaussian at $t=0$, that is

$$\varphi(x,0) = (2\pi\sigma^2)^{-1/4} \exp\left(-\frac{x^2}{4\sigma^2}\right). \tag{20}$$

The coefficient guarantees the normalization of φ. By axiom 1, we have

$$\mathbb{E}_{\varphi}[X](0) := \int_{\mathbb{R}} \mathrm{d}x\, \varphi^*(x,0)\varphi(x,0)x = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mathbb{R}^d} \mathrm{d}x \exp\left(-\frac{x^2}{2\sigma^2}\right) x = 0,$$

and

$$\mathrm{Var}_{\varphi}[X](0) := \int_{\mathbb{R}} \mathrm{d}x\, \varphi^*(x,0)\varphi(x,0)x^2 = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mathbb{R}^d} \mathrm{d}x \exp\left(-\frac{x^2}{2\sigma^2}\right) x^2 = \sigma^2.$$

Then, equation 17 gives $\Delta X = \sqrt{\mathrm{Var}_{\varphi}[X](0)} = \sigma$.

To evaluate $\Delta V(0)$, we have to use time evolution equation to calculate X at time Δt. By equation 10, we have (for brevity, we omit the subscript cut in N_{cut})

$$\varphi(x,\Delta t) = \varphi(x,0) - \mathrm{i}\Delta t \sum_{n=0}^{N} \frac{(-1)^n}{n!} R_n(x) \partial^n \varphi(x,0) - \frac{\mathrm{i}}{2}\Delta t^2 \sum_{n=0}^{N} \sum_{n'=0}^{N} \cdots.$$

We have to evaluate up to $o(\Delta t^2)$, so as to give the variance of velocity which scales as $x^2(\Delta t)/\Delta t^2$. But this would be too complicated (the double summation). For simplification, we consider a "free particle" where all R_ns except for R_N are vanishing. This is the situation when σ tends to zero, because there are more σ factors in the denominator if there are more derivatives on f, and as σ tends to zero, the term proportional to $R_N(x)\partial^N[f(x,0)]$ surpasses all the other terms.[7] In this situation, time evolution (equation 1) reduces to

$$\mathrm{i}\frac{\partial \varphi}{\partial t}(x,t) = \frac{(-1)^N}{N!} R_N(x) \partial^N \varphi(x,t).$$

7. In the traditional approach of quantum mechanics, $\sigma \to 0$ indicates that the momentum is large (since momentum is proportional to $\partial/\partial x$), so the term with the highest order of momentum will dominates the time evolution. When the momentum is large enough, the kinetic term dominates Hamiltonian, the potential becomes omittable, making the particle free (namely, unconstrained by potential).

Hence, up to $o(\Delta t^2)$ (we will see why $o(\Delta t)$ is insufficient),

$$\begin{aligned}\varphi(x,\Delta t) &= \varphi(x,0) + \frac{\partial\varphi}{\partial t}(x,0)\Delta t + \frac{1}{2}\frac{\partial^2\varphi}{\partial t^2}(x,0)\Delta t^2 + o(\Delta t^2)\\ &= \varphi(x,0) - \mathrm{i}\frac{(-1)^N}{N!}R_N(x)\partial^N\varphi(x,0)\Delta t - \frac{\mathrm{i}}{2}\frac{(-1)^N}{N!}R_N(x)\partial^N\frac{\partial\varphi}{\partial t}(x,0)\Delta t^2 + o(\Delta t^2).\end{aligned}$$

Inserting $(\partial\varphi/\partial t)$ and considering the surpassing again, we get

$$\varphi(x,\Delta t) = \varphi(x,0) - \mathrm{i}\frac{(-1)^N}{N!}R_N(x)\partial^N\varphi(x,0)\Delta t - \frac{1}{2}\left[\frac{(-1)^N}{N!}R_N(x)\right]^2\partial^{2N}\varphi(x,0)\Delta t^2 + o(\Delta t^2).$$

Once obtained $\varphi(x,\Delta t)$, we can calculate $\Delta V(0)$. First, we evaluate $V(0)$ as

$$V(0) = \lim_{\Delta t\to 0}\mathbb{E}_\varphi\left[\frac{X - \mathbb{E}_\varphi[X](0)}{\Delta t}\right](\Delta t) = \lim_{\Delta t\to 0}\frac{\mathbb{E}_\varphi[X](\Delta t) - \mathbb{E}_\varphi[X](0)}{\Delta t}.$$

Directly inserting $\varphi(x,\Delta t)$ results in

$$\begin{aligned}\mathbb{E}_\varphi[X](\Delta t) &= \int_{\mathbb{R}}\mathrm{d}x\,|\varphi(x,\Delta t)|^2 x\\ &= \mathbb{E}_\varphi[X](0) + \mathrm{i}\frac{(-1)^N}{N!}\int_{\mathbb{R}}\mathrm{d}x\varphi(x,0)\partial^N\varphi(x,0)[R_N^*(x) - R_N(x)]x\Delta t + o(\Delta t).\end{aligned}$$

Hence,

$$V(0) = \mathrm{i}\frac{(-1)^N}{N!}\int_{\mathbb{R}}\mathrm{d}x\varphi(x,0)\partial^N\varphi(x,0)[R_N^*(x) - R_N(x)]x.$$

When N is even, R_N is real (see section 8), we find $V(0) = 0$. But when N is odd, R_N is purely imaginary. In this situation, denote $R_N(x) = \mathrm{i}A(x)/2$, we get

$$V(0) = \frac{(-1)^N}{N!}\int_{\mathbb{R}}\mathrm{d}x\,\varphi(x,0)\partial^N\varphi(x,0)A(x)\,x,$$

which may not vanish, leading to a violation of parity symmetry, because the velocity has a favored direction even though the distribution (or wavefunction) has not. So, from now on, we only consider when N is even. For any random variable R, we have the formula

$$\mathrm{Var}[R] = \mathbb{E}[(R - \mathbb{E}[R])^2] = \mathbb{E}[R^2 - 2\mathbb{E}[R]R - \mathbb{E}^2[R]] = \mathbb{E}[R^2] - \mathbb{E}^2[R].$$

Applying to equation 19 gives

$$\Delta V^2(0) = \lim_{\Delta t\to 0}\mathbb{E}_\varphi\left[\left(\frac{X - \mathbb{E}_\varphi[X](0)}{\Delta t}\right)^2\right](\Delta t) - V^2(0) = \lim_{\Delta t\to 0}\frac{\mathbb{E}_\varphi[X^2](\Delta t)}{\Delta t^2},$$

where we have used $\mathbb{E}_\varphi[X](0) = V(0) = 0$ when N is even. Since R_N has been real, we have

$$\begin{aligned}\mathbb{E}_\varphi[X^2](\Delta t) &= \int_{\mathbb{R}}\mathrm{d}x\,\varphi^*(x,\Delta t)\varphi(x,\Delta t)x^2\\ &= \left(\frac{1}{N!}\right)^2\int_{\mathbb{R}}\mathrm{d}x\left[\partial^N\varphi(x,0)\partial^N\varphi(x,0) - \frac{1}{2}\varphi(x,0)\partial^{2N}\varphi(x,0)\right]R_N^2(x)x^2\Delta t^2 + o(\Delta t^2).\end{aligned}$$

So,

$$\Delta V^2(0) = \left(\frac{1}{N!}\right)^2\int_{\mathbb{R}}\mathrm{d}x\left[\partial^N\varphi(x,0)\partial^N\varphi(x,0) - \frac{1}{2}\varphi(x,0)\partial^{2N}\varphi(x,0)\right]R_N^2(x)x^2.$$

As we have expected, the terms that involve the partial derivatives of φ share the *same* (highest) order of $1/\sigma$, surpassing all the terms that we have neglected (which have lower order of $1/\sigma$) as σ tends to zero. Assume that R_N is entire, so that we can Taylor expand R_N^2 at origin, as

$$R_N^2(x) = R_N^2(0) + \partial R_N^2(0)x + \frac{1}{2}\partial^2 R_N^2(0)x^2 + \cdots.$$

We arrive at

$$\Delta V^2(0) = \left(\frac{1}{N!}\right)^2\sum_{n=0}^{\infty}\frac{\partial^n R_N^2(0)}{n!}\int_{\mathbb{R}}\mathrm{d}x\left[\partial^N\varphi(x,0)\partial^N\varphi(x,0) - \frac{1}{2}\varphi(x,0)\partial^{2N}\varphi(x,0)\right]x^{2+n}. \tag{21}$$

With the aid of computer, we evaluate the term in $\Delta V^2(0)$ for each N and n (see appendix C. To satisfy uncertainty principle (axiom 4), $\Delta V^2(0)$ shall be proportional to $1/\sigma^2$, so that $\Delta X \Delta V$ is independent of σ. Such terms are found when $(N, n) \in \{(2,0), (4,4), (6,8), (8,12), \ldots\}$, indicating a relation $n = 4(N/2 - 1) = 2N - 4$. So, we have $\partial^n R_N^2(0) = 0$ except for $n = 2N - 4$. In other words,

$$R_N(x) \propto x^{N-2}. \tag{22}$$

So, uncertainty principle restricts the highest order moment to be an explicit power function.

The specific case is $N = 2$ in which R_N becomes constant. This corresponds to the canonical situation in physics. To determine this constant, we have (set $N = 2$, $n = 0$)

$$\Delta V(0) = \frac{|R_2|}{2} \sqrt{\int_{\mathbb{R}} \mathrm{d}x \left[\partial^2 \varphi(x,0) \partial^2 \varphi(x,0) - \frac{1}{2} \varphi(x,0) \partial^4 \varphi(x,0)\right] x^2} = \frac{|R_2|}{2} \sqrt{\frac{23}{32\sigma^2}},$$

where the integral is evaluated by computer. Then, by uncertainty principle, $\Delta X(0) \Delta V(0) = \sqrt{23/32} |R_2|/2 \sim \hbar/(2m)$, implying

$$|R_2| \sim \sqrt{\frac{32}{23}} \frac{\hbar}{m} \approx 1.18 \frac{\hbar}{m}. \tag{23}$$

This is consistent with the traditional Hamiltonian where $R_2 = -\hbar/m$.

We can generalize the previous analysis to dimension $d > 1$, in which the covariance matrix of f is diagonal, such that all dimensions are independent. Namely,

$$\varphi(x, 0) = \prod_{\alpha=1}^{d} (2\pi(\sigma^\alpha)^2)^{-1/4} \exp\left(-\frac{1}{4}\left(\frac{x^\alpha}{\sigma^\alpha}\right)^2\right).$$

The previous analysis, then, is taken on each dimension individually, resulting in exactly the same result for each dimension.

10 From Quantum to Classical (TODO)

Appendix A

Taylor Reminder

Given a function $f\in\mathcal{C}^n(\mathbb{R}^d)$, we are to calculate the reminder $h_n(x,x_0)$ in the Taylor expansion at x_0

$$f(x)=f(x_0)+\sum_{k=1}^{n}\frac{1}{k!}(\partial_{\alpha_1}\cdots\partial_{\alpha_k}f)(x_0)\prod_{i=1}^{k}(x^{\alpha_i}-x_0^{\alpha_i})+h_{n+1}(x,x_0).$$

We begin at a variation of the fundamental theorem of calculus

$$f(x)-f(x_0)=\int_0^1\mathrm{d}t\,(\partial_{\alpha_1}f)(x+t(x_0-x))(x^{\alpha_1}-x_0^{\alpha_1}).$$

Integration by parts gives

$$f(x)-f(x_0)=(\partial_{\alpha_1}f)(x_0)(x^{\alpha_1}-x_0^{\alpha_1})+\frac{1}{2}\int_0^1\mathrm{d}(t^2)\,(\partial_{\alpha_1}\partial_{\alpha_2}f)(x+t(x_0-x))(x^{\alpha_1}-x_0^{\alpha_1})(x^{\alpha_2}-x_0^{\alpha_2}).$$

Applying integration by parts on the integral again, we get

$$\begin{aligned}f(x)-f(x_0)=\ &(\partial_{\alpha_1}f)(x_0)(x^{\alpha_1}-x_0^{\alpha_1})\\&+\frac{1}{2!}(\partial_{\alpha_1}\partial_{\alpha_2}f)(x_0)(x^{\alpha_1}-x_0^{\alpha_1})(x^{\alpha_2}-x_0^{\alpha_2})\\&+\frac{1}{3!}\int_0^1\mathrm{d}(t^3)\ (\partial_{\alpha_1}\partial_{\alpha_2}\partial_{\alpha_3}f)(x+t(x_0-x))(x^{\alpha_1}-x_0^{\alpha_1})(x^{\alpha_2}-x_0^{\alpha_2})(x^{\alpha_3}-x_0^{\alpha_3}).\end{aligned}$$

Repeating this process gives

$$\begin{aligned}f(x)-f(x_0)=\ &(\partial_{\alpha_1}f)(x_0)(x^{\alpha_1}-x_0^{\alpha_1})\\&+\frac{1}{2!}(\partial_{\alpha_1}\partial_{\alpha_2}f)(x^{\alpha_1}-x_0^{\alpha_1})(x^{\alpha_2}-x_0^{\alpha_2})\\&+\cdots\\&+\frac{1}{n!}\int_0^1\mathrm{d}(t^n)\,(\partial_{\alpha_1}\cdots\partial_{\alpha_n}f)(x+t(x_0-x))(x^{\alpha_1}-x_0^{\alpha_1})\cdots(x^{\alpha_n}-x_0^{\alpha_n}).\end{aligned}$$

So, the Taylor reminder reads

$$h_n(x,x_0)=\frac{1}{n!}\prod_{i=1}^{n}(x^{\alpha_i}-x_0^{\alpha_i})\times\int_0^1\mathrm{d}(t^n)\,(\partial_{\alpha_1}\cdots\partial_{\alpha_n}f)(x+t(x_0-x)).\tag{A.1}$$

Appendix B

Rapidly Decreasing Entire Functions are Dense in Square-Integrable Space (TODO)

A function $f: \mathbb{R}^d \to \mathbb{C}$ is square-integrable if

$$\int_{\mathbb{R}^d} \mathrm{d}x |f(x)|^2 := \int_{\mathbb{R}^d} \mathrm{d}x f^*(x) f(x) < \infty.$$

This is a improper integral, thus defined by

$$\int_{\mathbb{R}^d} \mathrm{d}x |f(x)|^2 = \lim_{R \to \infty} \int_{B(R)} \mathrm{d}x |f(x)|^2,$$

where $B(R) := [-R, R]^d$ denotes the d-dimensional “box”. In other words, for any $\varepsilon > 0$, there exists $R_\star > 0$ and $\delta > 0$, such that for any $R > R_\star$ and $0 < \Delta x < \delta$,

$$\left| \int_{\mathbb{R}^d} \mathrm{d}x |f(x)|^2 - \int_{B(R)} \mathrm{d}x |f(x)|^2 \right| < \varepsilon.$$

So, for any $\varepsilon > 0$, we can construct a compact supported function $g_R: \mathbb{R}^d \to \mathbb{C}$, which consists with f within the box $B(R)$ and vanishes outside. TODO

Consider the convolution

$$f_n(x) = (\delta_n * f)(x) := \int_{\mathbb{R}^d} \mathrm{d}y \delta_n(x - y) f(y),$$

where n is a positive integer and

$$\delta_n(x) := \frac{1}{(\sqrt{2\pi} n)^d} \exp\left(-\frac{x^2}{2n}\right),$$

that is, the probabilistic density function of normal distribution with zero mean and variance n. By young’s inequality for convolution, we have

$$\|\delta_n * f\|_\infty \leqslant \|\delta_n\|_2 \|f\|_2.$$

Appendix C
Computing the Uncertainty of Velocity

We use Wolfram Mathematica to compute the integral in equation 21. Namely, given (N, n), we are to evaluate

$$\int_{\mathbb{R}} \mathrm{d}x \left[\partial^N \varphi(x,0) \partial^N \varphi(x,0) - \frac{1}{2} \varphi(x,0) \partial^{2N} \varphi(x,0) \right] x^{2+n},$$

where φ is given by equation 20. In Mathematica, we first input φ (re-denoted by f) as

```
f[x_, sigma_] := (2*Pi*sigma^2)^(-1/4)*Exp[-x^2/4/sigma^2];
```

Then, input the target integral as

```
targetIntegral[sigma_, N_, n_] := Simplify[Integrate[(D[f[x, sigma], {x,
N}]^2 - f[x, sigma]* D[f[x, sigma], {x, 2*N}] / 2) * x^(n+2), {x, -Infinity,
Infinity}, Assumptions->{Element[sigma, Reals], sigma > 0}]];
```

where we have to assert that `sigma` is a positive real number. Then, we try (we have to re-denote N by `m` because `N` has been a keyword of Mathematica)

```
list = Flatten[Table[{{m, n}, targetIntegral[sigma, m, n]}, {m, {2, 4, 6, 8,
10}}, {n, 0, 30}], 1]
```

which outputs a list of (N, n) pairs and the corresponding integrals, each element has the format `{{N,n}, integral}`. To filter the valid items, in which the integral is proportional to $1/\sigma^2$, we append

```
result = Select[list, MatchQ[#[[2]], _*sigma^-2] &]
```

The `MatchQ` function matches the integrals (as the second element) that are proportional to $1/\sigma^2$.